Stephanie M Noble

ASSISTANT PROFESSOR · COMPUTATIONAL NEUROSCIENCE

621 Interdisciplinary Science & Engineering Complex | Boston MA 02120

s.noble@northeastern.edu | 860 416 2384 | neuroprismlab.github.io | @sNeuroble | @sNeuroble

Academic Appointments

Assistant Professor, Northeastern University

Boston MA

DEPARTMENT OF PSYCHOLOGY
DEPARTMENT OF BIOENGINEERING
INSTITUTE FOR COGNITIVE AND BRAIN HEALTH

July 2023 – present

Research Affiliate, Yale University

New Haven CT

RADIOLOGY & BIOMEDICAL IMAGING

July 2023 – present

Postdoctoral Associate, Yale University

New Haven CT

RADIOLOGY & BIOMEDICAL IMAGING

Aug. 2019 – June 2023

- Advisor: Dustin Scheinost

Education

PhD, Yale University

New Haven CT

INTERDEPARTMENTAL NEUROSCIENCE PROGRAM (INP)

Sept. 2014 – May 2019

- Dissertation: Reliability & Validity of fMRI Mapping Methods (Qualified for Candidacy with Distinction)
- Advisor: R. Todd Constable

BSE, Princeton University

Princeton NJ

CHEMICAL & BIOLOGICAL ENGINEERING: BIOTECHNOLOGY & BIOINFORMATICS TRACK
HONORS CERTIFICATE IN QUANTITATIVE & COMPUTATIONAL NEUROSCIENCE
CERTIFICATE IN ENGINEERING BIOLOGY

Sept. 2008 – May 2012

- Thesis: Muscle Contraction as a Markov Process
- Advisor: Clarence E. Schutt

Experience

Elite Warrior Identification, LLC

Arlington VA

INDEPENDENT CONSULTANT

Mar. 2022 – Dec 2024

- EEG connectivity analysis and machine learning

Source Signal Imaging, LLC

San Diego CA

INDEPENDENT CONSULTANT

Oct. 2013 – Aug 2014

- Research and prototyping for EEG source estimation projects

goBlue Labs, LLC

New Haven CT

FOUNDING CHIEF SCIENCE OFFICER (CSO)

2012 – 2013

- Real-time EEG source estimation and neurofeedback software

Grants

ONGOING

2023 – 2027 (NCE)

NIH BRAIN Initiative K99/R00 Award (R00MH130894)
National Institute of Mental Health
"Empirical Power Analysis Tool for fMRI"
Amount: $249,000
Role: PI

2026 –

1Kd Award
Wellcome Leap
"Early-life Gut Microbiome Resilience as a Foundation for Neurodevelopmental Adaptation"
Amount: $2,392,506
Role: Co-I (PI: Laurel Gabard-Durnam)

PENDING

Submitted 2025

Functional Communication Frontier Science Hub Anchor Project
Aligning Research to Impact Autism
Amount: Pending
Role: Co-I, "Computational Neuroimaging Lead" (PIs: Kristina Johnson & Meghan Swanson)

COMPLETED

2024 - 2025

CBH Seed Grant
Northeastern University Institute for Cognitive & Brain Health
"A Precision Medicine Framework for Predicting Future Suicidal Ideation in Adolescents"
Amount: $75,000
Role: Co-PI (with Joshua Curtiss; Co-I Susan Whitfield-Gabrieli)

2022 - 2023

NIH BRAIN Initiative K99/R00 Award (K99MH130894)
National Institute of Mental Health
"Empirical Power Analysis Tool for fMRI"
Amount: $122,677
Role: PI

2019 - 2022

NIH Blueprint DSPAN Award, F99/K00 (K00MH122372)
National Institute of Mental Health
"Constrained Network-Based Multiple Comparison Correction"
Amount: $241,540
Role: PI

2018 - 2019

NIH Blueprint DSPAN Award, F99/K00 (F99NS108557)
National Institute of Neurological Disorders and Stroke
Improving Reliability and Validity of fMRI Statistical Methods
Amount: $43,497
Role: PI

2016 - 2018

Graduate Research Fellowship Program (DGE1122492)
National Science Foundation
Amount: $92,000
Role: PI

Honors & Awards

Research

2025	**Mutual Mentoring Advancement Program (M2AP)**, Northeastern University, $3,000
2025	**Early-Career Rigor Champions Prize**, National Institute of Neurological Disorders and Stroke, $10,000
2024, 2025	**Full Member Nomination**, Sigma Xi
2019	**Merit Abstract Award,** Organization for Human Brain Mapping, $2,000 (15 awardees)
2019	**Associate Member Nomination**, Sigma Xi
2018 - 2019	**Program for Excellence in Science Fellowship**, AAAS / Science
2018	**Annie Le Fellowship**, Yale University (stipend & professional enrichment supplement; academic excellence and service to the community)
2017	**Qualified for Candidacy with Distinction**
2016	**Best Poster Award**, Yale Biomedical Engineering Retreat
2015 - 2017	**Neuroscience Scholars Program Fellowship**, Society for Neuroscience (15 awardees, support for society meeting attendance, society membership, professional enrichment funds)
2012	**Honors Certification in Quantitative & Computational Neuroscience**
2010	**Siebel Energy Grand Challenges Fellowship**, Princeton University, $4,500
2009 - 2012	**Howard Hunt Garmany Memorial Scholarship**, Hartford Foundation for Public Giving (awarded annually)

Outreach

2016	**WE16 Outreach Award**, Society of Women Engineers (to Yale GradSWE; outreach co-chair)
2016	**Seton Elm-Ivy Award**, The Community Foundation for Greater New Haven (to INP Outreach; co-chair)

Industry

2013	**Innovation Fund Award**, Yale Entrepreneurial Institute, $100,000 (offered) (exclusive award to Yale start-up)
2012	**TechStart Accelerator Program Fund Award,** Connecticut Innovations, $25,000 (exclusive award to 5 CT start-ups)
2012	**Private Investment**, Bridge Builders Collaborative, undisclosed

Publications

H-index=30, Accepted=58, First Author=9, Last Author=4, Google Scholar: https://scholar.google.com/citations?user=JxQdvn4AAAAJ

** = all authors contributed equally*

Accepted

1. Doshvargar, F., Cravo, F., Shearer, H., **Noble, S.** In press. Mass univariate aggregation methods for machine learning in neuroscience. In press. *Aperture Neuro*.
2. Zainal, H.N., **Noble, S.** In press. Blunted Diurnal Cortisol Slope as a Risk Marker for 9-Year Executive Function Decline: Moderation by Daily Stress and Generalized Anxiety Disorder Severity. In Press. *Journal of Gerontology: Psychological Sciences*.
3. Adkinson, B.D., Rosenblatt, M., Sun, H., Dadashkarimi, J., Tejavibulya, L., Horien, C., Westwater, M.L., **Noble, S.**, Scheinost, D. 2026. Feature selection leads to divergent neurobiological interpretations of brain-based machine learning biomarkers. *Nature Human Behavior*. (Preprint: https://www.biorxiv.org/content/10.1101/2025.03.12.642878.full.pdf)
4. Shearer, H., Rosenblatt, M., Ye, J., Jiang, R., Tejavibulya, L., Foster, M., … Nichols, T., Curtiss, J., Scheinost, D., **Noble, S**. 2025. BrainEffeX: A Web App for Exploring fMRI Effect Sizes. *Aperture Neuro.* (Preprint: https://osf.io/kryn4_v1)
5. Rodriguez, R.X., **Noble, S.,** Camp, C., Scheinost, D. 2025. Connectome caricatures remove large-amplitude co-activation patterns in resting-state fMRI to emphasize individual differences. *Nature Neuroscience.* (Preprint: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11030410/pdf/nihpp-2024.04.08.588578v1.pdf)
6. Xu, T., Fischbach, A., Bridgeford, E.W., Bayrak, R.G., Vogelstein, J., **Noble, S.,** Anteraper, S. In press. Book Chapter in: Current limitations in functional connectivity assessment: suggestions for analysis improvements. In M.A. Rocca & M. Filippi (Eds.), Functional connectivity of the human brain: from mechanisms to clinical applications. Academic Press.
7. O'Neill, A. G., Pax, M., Sepulcre, J., Camprodon, J., Holt, D. J., **Noble, S.**, Roffman, J. L., Eryilmaz, H. 2025. Predicting cognitive functioning in early psychosis: factors supporting and limiting generalizability of connectome-based models. *NPP—Digital Psychiatry and Neuroscience*.
8. Bridgeford, E. W., Powell, M., Kiar, G., **Noble, S.**, Chung, J., Panda, S., Lawrence, R., Priebe, C.E., Caffo, B., Xu, T., Milham, M., Vogelstein, J. T. 2025. When no answer is better than a wrong answer: a causal perspective on batch effects. *Imaging Neuroscience*. (Preprint: https://www.biorxiv.org/content/10.1101/2021.09.03.458920v4)
9. Shearer, H., Eilbott, J., Vila-Rodriguez, F., Xu, T., **Noble, S.,** Vanderwal, T. 2024. Comparing reliability-based measures of functional connectivity between movie and rest: an ROI-based approach. *Imaging Neuroscience*.
10. Adkinson, B.D., Rosenblatt, M., Dadashkarimi, J., Tejavibulya, L., Jiang, R., **Noble, S.,** Scheinost, D. 2024. Brain-phenotype predictions of language and executive function can survive across diverse real-world data: Dataset shifts in developmental populations. *Developmental Cognitive Neuroscience*.
11. Rosenblatt, M.., Tejavibulya, L., Camp, C., Jiang, R., Westwater, M., **Noble, S.**, Scheinost, D. 2024. Power and reproducibility in the external validation of brain-phenotype predictions. *Nature Human Behavior*.
12. Rosenblatt, M., Mehta, S., Peterson, H., Dadashkarimi, J., Rodriguez, R.X., Foster, M.L., Adkinson, B.D., Liang, Q., Kimble, V.M., Ye, J., McCusker, M.C., Farruggia, M.C., Rolison, M., Westwater, M.L., Jiang, R., **Noble, S.,** Scheinost, D. 2024. Trends in self-citation rates in neuroscience literature. *eLife*. (Preprint: https://www.biorxiv.org/content/10.1101/2022.09.27.509533v1)
13. Gell, M., **Noble, S.,** Laumann, T.O., Nelson, S.M., Tervo-Clemmens, B. 2024. Psychiatric Neuroimaging Designs for Individualised, Cohort, and Population Studies. Accepted. *Neuropsychopharmology*. (Preprint: https://osf.io/278ef/)
14. Kam, J., Badhwar, A., Borghesani, V., Lee, K., **Noble, S.,** Raamana, P. R., ... & Tzovara, A. 2024. Creating Diverse and Inclusive Scientific Practices for Research Datasets and Dissemination. *Imaging Neuroscience*. (Preprint: https://osf.io/preprints/psyarxiv/dr5hg)
15. Jiang, R., **Noble, S.,** Rosenblatt, M., Dai, W., Ye, J., Liu, S., Qi, S., Calhoun, V.D., Sui, J., Scheinost, D. 2024. The brain structure, inflammatory, and genetic mechanisms underlying the association between physical frailty and depression. *Nature Communications*.
16. **Noble, S.*,** Curtiss, J.*, Pessoa, L., Scheinost, D. 2024. The tip of the iceberg: a call to embrace anti-localizationism in human neuroscience research. *Imaging Neuroscience.* (Preprint: https://osf.io/preprints/psyarxiv/9eqh6)

17. Mansour, S., Seguin, C., Winkler, A., **Noble, S.,** Zalesky, A. Topological Cluster Statistic (TCS): Towards structural-connectivity-guided fMRI cluster enhancement. 2024. *Network Neuroscience*. (Preprint: https://www.researchsquare.com/article/rs-2059418/v1)

18. Camp, C.C., **Noble, S.**, Scheinost, D, Stringaris, A., Nielson, D.M. 2024. Test-retest reliability of functional connectivity in depressed adolescents. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging.* (Preprint: https://www.medrxiv.org/content/10.1101/2022.10.11.22280962v1)

19. Rosenblatt, M.., Tejavibulya, L., Jiang, R., **Noble, S.**, Scheinost, D. 2024. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications.* (Preprint: https://osf.io/ptuwe.)

20. Sun, H., Jiang, R., Dai, W., Dufford, A.J., **Noble, S.,** Gu, S., Spann, M., Scheinost, D. 2023. Network controllability of structural connectomes in the neonatal brain. *Nature Communications.*

21. Dadashkarimi, J., Karbasi, A. Liang, Q., Rosenblatt, M., **Noble, S.**, Foster, M., Rodriguez, R., Adkinson, B., Ye, J., Sun, H., Camp, C., Farruggia, M., Tejavibulya, L., Dai, W., Jiang, R., Pollatou, A., Scheinost, D. 2023. Cross Atlas Remapping via Optimal Transport (CAROT): Creating connectomes for any atlas when raw data is not available. *Medical Image Analysis (MEDIA).* (Preprint: https://www.biorxiv.org/content/10.1101/2022.07.19.500642v2)

22. Ye, J., Sun, H., Gao, S., Dadashkarimi, J., Rosenblatt, M., Rodriguez, R.X., Mehta, S., Jiang, R., **Noble, S.**, Westwater, M.L., Scheinost, D., 2023. Altered Brain Dynamics across Bipolar Disorder and Schizophrenia during Rest and Task-switching Revealed by Overlapping Brain States. *Biological Psychiatry.* (Preprint: https://www.medrxiv.org/content/10.1101/2022.10.07.22280835v1)

23. Rosenblatt, M.., Rodriguez, R., Westwater, M. Horien, C., Greene, A., Constable, R.T., **Noble, S.**, Scheinost, D., 2023. Connectome-based machine learning models are vulnerable to subtle data manipulations. *Cell Patterns.* (Preprint: https://osf.io/ptuwe)

24. Jiang, R., **Noble, S.,** Sui, J., Yoo, K., Rosenblatt, M., Horien, C., Qi, S., Liang, Q., Sun, H., Calhoun, V.D., Scheinost, D. 2023. Associations of physical frailty with health outcomes and brain structure in 483,033 middle-aged and older adults from the UK Biobank. *The Lancet Digital Health*.

25. Shinn, M., Hu, A., Turner, L., **Noble, S.**, Achard, S., Anticevic, A., Scheinost, D., Constable, R.T., Lee, D., Bullmore, E.T., Murray, J.D. 2023. Functional brain networks reflect spatial and temporal autocorrelation. *Nature Neuroscience*. (Preprint: https://www.biorxiv.org/content/10.1101/2021.06.01.446561v1)

26. Yang, G., Bozek, J., **Noble, S.**, Han, M., Wu, X., Xue, M., Kang, J., Jia, T., Fu, J., Ge, J., Cui, Z., Li, X., Feng, J., Gao, J. 2023. Global diversity in individualized cortical network topography. *Cerebral Cortex*.

27. Jiang, R., Calhoun, V.D., **Noble, S.,** Sui, J., Liang, Q., Qi, S., Scheinost, D. 2023. A functional connectome signature of blood pressure in> 30 000 participants from the UK biobank. *Cardiovascular Research*.

28. Scheinost, D., Pollatou, A., Dufford, A.J., Jiang, R., Farruggia, M.C., Rosenblatt, M., Peterson, H., Rodriguez, R.X., Dadashkarimi, J., Liang, Q., Dai, W., Foster, M.L., Camp, C.C., Tejavibulya, L., Adkinson, B.D., Sun, H., Ye, J., Cheng, Q., Spann, M.N., Rolison, M., **Noble, S.***, Westwater, M.L.* 2023. Machine learning and prediction in fetal, infant, and toddler neuroimaging: a review and primer. *Biological Psychiatry*.

29. Rodriguez, R., **Noble, S.**, Tejavibulya, L., Scheinost, D. 2022. Leveraging edge-centric networks complements existing network-level inference for functional connectomes. *NeuroImage*.

30. Dai, W., **Noble, S**., & Scheinost, D. 2022. The Semi-constrained Network-Based Statistic (scNBS): Integrating Local and Global Information for Brain Network Inference. *Medical Image Computing and Computer Assisted Intervention (MICCAI).*

31. Jiang, R., Westwater, M.L., **Noble, S.,** Rosenblatt, M., Dai, W., Qi, S., Sui, J., Calhoun, V.D., Scheinost, D. 2022. Associations between grip strength, brain structure, and mental health in> 40,000 participants from the UK Biobank. *BMC Medicine*.

32. **Noble, S.,** Mejia, M., Zalesky, A., Scheinost, D. 2022. Improving power in functional magnetic resonance imaging by moving beyond cluster-level inference. *Proceedings of the National Academy of Sciences*. (Preprint: https://www.biorxiv.org/content/10.1101/2021.09.23.461354v1)

33. Greene, A.S., Shen, X.., **Noble, S.**, Hahn, A., Arora, J., Tokoglu, F., Spann, M.N., Barron, D.S., Sanacora, G., Srihari, V.H., Woods, S.W., Scheinost, D., Constable, R.T. 2022. Individuals who defy stereotypical profiles require distinct brain-phenotype models. *Nature*.

34. Tejavibulya, L., Rolison, M., Gao, S., Liang, Q. Peterson, H., Dadashkarimi, J., Farruggia, M., Hahn, A.C., **Noble, S.**, Lichenstein, S.D., Pollatou, A., Dufford, A.J., Scheinost, D. 2022. Predicting the future of neuroimaging predictive models in psychiatry. *Molecular Psychiatry*.

35. Horien, C., Floris, D.L., Greene, A.S., **Noble, S.**, Rolison, M., Tejavibulya, L., O'Connor, D., McPartland, J.C., Scheinost, D., Chawarska , K., Lake, E.M., Constable, R.T. 2022. Functional connectome-based predictive modelling in autism. *Biological Psychiatry*.

36. Tejavibulya, L., Peterson, H., Greene, A., Gao, S., Rolison, M., **Noble, S.**, Scheinost, D. 2022. Large-scale differences in functional organization of left- and right-handed individuals using whole-brain, data-driven analysis of connectivity. NeuroImage.

37. Horien, C., Lee, K., Westwater, M., **Noble, S.**, Tejavibulya, L., Kayani, T., Constable, R.T., Scheinost, D. 2021. A protocol for working with open-source neuroimaging datasets. *STAR Protocols*.

38. Dufford, A.J., **Noble, S.**, Gao, S., Scheinost, D. 2022. The instability of functional connectomes across the first year of life. *Developmental Cognitive Neuroscience*. (Preprint: https://doi.org/10.1101/2021.04.14.439877)

39. Ibrahim, K., **Noble, S.**, He, G., Lacadie, C., Crowley, M.J., McCarthy, G., Scheinost, D., and Sukhodolsky, D.G. 2021. Large-Scale Functional Brain Networks of Maladaptive Childhood Aggression Identified by Connectome-Based Predictive Modeling. *Molecular Psychiatry*.

40. Bridgeford, E. W., Wang, S., Yang, Z., Wang, Z., Xu, T., Craddock, C., ... **Noble, S.**, Priebe, C.E., Caffo, B., Milham, M., Zuo, X., Consortium for Reliability and Reproducibility, Vogelstein, J. T. 2021. Eliminating accidental deviations to minimize generalization error and maximize reliability: applications in connectomics and genomics. *PLOS Computational Biology*. (Preprint: https://www.biorxiv.org/content/10.1101/802629v7)

41. Levitis, E., Gould van Praag, C. D., Gau, R., Heunis, S., DuPre, E., Kiar, G., … **Noble, S.**, … Maumet, C. 2021. Centering inclusivity in the design of online conferences. *Gigascience*. (Preprint: https://doi.org/10.31234/osf.io/vj5tu)

42. **Noble, S.**, Scheinost, D., Constable, R.T., 2021. A guide to the measurement and interpretation of fMRI test-retest reliability. *Current Opinion in Behavioral Sciences*. (*Invited Review, Deep Imaging Special Issue*).

43. Gau, R.*, **Noble, S.*,** Heuer, K.*, Bottenhorn, K.*, Bilgin, I.P.*, Yang, Y.*, Huntenburg, J.*, Bayer, J.M.M.*, Bethlehem, R.*, … Brainhack community. 2021. Brainhack: developing a culture of open, inclusive, community-driven neuroscience. *Neuron*. (Preprint: https://psyarxiv.com/rytjq/)

44. Barron, D.S., Gao, S., Dadashkarimi, J., Greene, A.S., Spann, M.N., **Noble, S.**, Lake, E., Krystal, J.H., Constable, R.T., Scheinost, D., 2020. Transdiagnostic, Connectome-Based Prediction of Memory Constructs Across Psychiatric Disorders. *Cerebral Cortex*. (Preprint: https://www.biorxiv.org/content/10.1101/638825v1)

45. Horien, C., **Noble, S.**, Greene, A.S., Lee, K., Barron, D.S., Gao, S., O'Connor, D., Salehi, M., Dadashkarimi, J., Shen, X., Lake, E.M., Constable, R.T., Scheinost, D., 2020. A Hitchhiker's Guide to Working with Large, Open-Source Neuroimaging Datasets. *Nature Human Behavior*.

46. **Noble, S.**, Scheinost, D., 2020. The Constrained Network-Based Statistic: A New Level of Inference for Neuroimaging. *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Proceedings, Part VII 23, 458-468.

47. Greene, A.S., Gao, S., **Noble, S.**, Scheinost, D., Constable, R.T., 2020. How Tasks Change Whole-Brain Functional Organization to Reveal Brain-Phenotype Relationships. *Cell Reports* 32, 108066.

48. **Noble, S.**, Scheinost, D., & Constable, R. T., 2020. Cluster failure or power failure? Evaluating sensitivity in cluster-level inference. *NeuroImage* 209, 116468.

49. **Noble, S.**, Scheinost, D., Constable, R.T., 2019. A decade of test-retest reliability of functional connectivity: A systematic review and meta-analysis. *NeuroImage* 203, 116157.

50. Dadashkarimi, J., Gao, S., Yeagle, E., **Noble, S.**, Scheinost, D., 2019. A Mass Multivariate Edge-wise Approach for Combining Multiple Connectomes to Improve the Detection of Group Differences. *International Workshop on Connectomics in Neuroimaging*. Springer, Cham, 64-73.

51. Yoo, K., Rosenberg, M.D., **Noble, S.**, Scheinost, D., Constable, R.T., Chun, M.M., 2019. Multivariate approaches improve the reliability and validity of functional connectivity and prediction of individual behaviors. *NeuroImage* 197, 212-223.

52. Scheinost, D., **Noble, S.**, Horien, C., Greene, A.S., Lake, E.M., Salehi, M., Gao, S., Shen, X., O'Connor, D., Barron, D.S., Yip SW., Rosenberg, M.D., Constable, R.T., 2019. Ten simple rules for predictive modeling of individual differences in neuroimaging. *NeuroImage*.
53. Lake, E.M., Finn, E.S., **Noble, S.M.**, Vanderwal, T., Shen, X., Rosenberg, M.D., Spann, M.N., Chun, M.M., Scheinost, D., Constable, R.T., 2019. The Functional Brain Organization of an Individual Allows Prediction of Measures of Social Abilities Transdiagnostically in Autism and Attention-Deficit/Hyperactivity Disorder. *Biological Psychiatry*.
54. Horien, C., **Noble, S.**, Finn, E.S., Shen, X., Scheinost, D., Constable, R.T., 2018. Considering factors affecting the connectome-based identification process: Comment on Waller et al. *NeuroImage* 169, 172-175.
55. **Noble, S.**, Spann, M.N., Tokoglu, F., Shen, X., Constable, R.T., Scheinost, D., 2017a. Influences on the test–retest reliability of functional connectivity MRI and its relationship with behavioral utility. *Cerebral Cortex* 27, 5415-5429.
56. **Noble, S.**, Scheinost, D., Finn, E.S., Shen, X., Papademetris, X., McEwen, S.C., Bearden, C.E., Addington, J., Goodyear, B., … Cannon, T.D., Constable, R.T., 2017b. Multisite reliability of MR-based functional connectivity. *NeuroImage* 146, 959-970.
57. Benjamin, C.F., Walshaw, P.D., Hale, K., Gaillard, W.D., Baxter, L.C., Berl, M.M., Polczynska, M., **Noble, S.**, Alkawadri, R., Hirsch, L.J., Constable, R.T., Bookheimer, S.Y., 2017. Presurgical language fMRI: mapping of six critical regions. *Human Brain Mapping* 38, 4239-4255.
58. Scheinost, D., Tokoglu, F., Shen, X., Finn, E.S., **Noble, S.**, Papademetris, X., Constable, R.T., 2016. Fluctuations in global brain activity are associated with changes in whole-brain connectivity of functional networks. *IEEE Transactions on Biomedical Engineering* 63, 2540-2549.

Under Review

1. **Noble, S.*,** Curtiss, J.*, Fischbach, A., Xu, T. Measurement & modeling tradeoffs for precision neuroimaging in psychiatry.
2. Cravo, F., **Noble, S.** Improving efficiency with the Incremental Cluster Threshold-Free Cluster Enhancement.
3. **Noble, S.,** Shearer, H., Rosenblatt, M., Ye, J., … Mejia, A., Nichols, T., Curtiss, J., Scheinost, D. Effect sizes in human functional neuroimaging.
4. Borovykh, A., Weissenbacher, M., **Noble, S.,** Shinn, M. A low-variance subspace underlies individual differences in resting state fMRI.
5. Christensen, E., Wiersch, L., Wang, M., McPherson, B., Shearer, H., **Noble, S.**, Lam, K., & *Poline., J.B., *Dhamala, E. Effect Sizes for Brain-Based Predictive Models of Cognition Across the Lifespan.
6. Cravo, F., Fischbach, A., Shearer, H., Rosenblatt, M., Scheinost, D., **Noble, S**. PRISME: A MATLAB Toolbox for Large Data-Driven Multimodal Power Benchmarking.
7. España-Irla, G., Tinney, E.M., Perko, M., Nwakamma, M., **Noble, S.,** Morris, T.P. Traumatic Brain Injury Associated with Altered Corpus Callosum Microstructure in Females: Exploring the Roles of Menopause Timing and Hormone Therapy in UK Biobank.

In Preparation

1. Merritt, H., Gonzalez, M.Z., **Noble, S.,** Teixeira, A.S. Dynamic social envirotypes disentangle divergent mental health outcomes and dissociable brain network organization
2. Curtiss, J.*, **Noble, S*.** A formalization of the IdioNomo Framework for explanation and prediction in psychology: unifying idiographic and nomothetic approaches (preprint: https://osf.io/vrb5a_v2).
3. **Noble, S.,** Shearer, H., Bray, S., Xu, T., Vanderwal, T. Reliability and Data Quantity. Chapter in White Paper: Best practices when using movies and story-listening to study neural responses with fMRI.
4. Fischbach, A., Shearer, H., Handwerker, D., Lewis, L., Quigley, K., Theriault, J., Satpute, A.B., Barrett, L.F., **Noble, S.** Deconfounding Subcortical Signals in fMRI by Modeling Region-Specific Cerebrospinal Fluid Noise.

5. Karamzadeh, M., Fischbach, A., **Noble, S.**, Loth, F. Machine Learning Classification of Asymptomatic Pediatric Chiari I Malformation from Resting-State fMRI.

Select Media Coverage

6. Lloyd, N. 2026. How a 'caricature' of a brain scan could lead to better treatment outcomes. *Northeastern Global News*. https://news.northeastern.edu/2026/01/05/functional-mri-breakthrough/
7. Rix, K. 2025. Precise maps of the brain's deepest corners are made possible through tools developed by these Northeastern researchers. *Northeastern Global News*. https://news.northeastern.edu/2025/09/02/northeastern-neuroprism-tools-map-brain/
8. Jahn, A. 2024. Andy's BrainCast #002: Stephanie Noble. *Andy's BrainCast*. https://www.youtube.com/watch?v=ALmwcecw7o8
9. McMurray, C. 2024. Data access changes to UK Biobank stir unease in neuroscientists. *The Transmitter*. https://www.thetransmitter.org/community/data-access-changes-to-uk-biobank-stir-unease-in-neuroscientists/
10. Lloyd, N. 2023. Psychology professor building 'data science tool' to increase the reliability of human brain research. *Northeastern Global News*. https://cos.northeastern.edu/news/psychology-professor-building-data-science-tool-to-increase-the-reliability-of-human-brain-research/
11. Locklear, M. 2022. To better understand the brain, look at the bigger picture. *YaleNews*. https://news.yale.edu/2022/08/04/better-understand-brain-look-bigger-picture
12. Yu, A. 2021. Scientists have used fMRI to study brain activity for years. Now, some question the results' reliability. *The Pulse. WHYY PBS NPR*. https://whyy.org/segments/scientists-used-fmri-to-study-brain-activity-for-years-now-some-question-the-results-reliability/
13. Proff, I., 2020. Can brain scans transform psychiatry? *Medium*. https://medium.com/@irisproff/can-brain-scans-transform-psychiatry-963ff2e5fb4f
14. Macmillon, T., 2012. Start-Up seeks to tap mind power. *New Haven Independent*. https://www.newhavenindependent.org/index.php/archives/entry/start-up_tries_to_tap_mind_power/

Presentations

Invited Seminars & Colloquia

2025 *ReproNim Webinar,* University of Massachusetts Chan Medical School

2025 *Northwestern University Interdepartmental Neuroscience Student Hosted Seminar Series,* Northwestern University

2024 *UCLA/BRI Neuroimaging Affinity Group Distinguished Colloquium Speaker,* University of California Los Angeles

2024 *Life Sciences Day,* Northeastern University

2023 *Developmental Cognition & Neuroimaging Lab, Masonic Institute for the Developing Brain,* University of Minnesota

2023 *COBRE Behavior and Neurodata Core,* Brown University

2022 *Department of Electrical and Computer Engineering* and *Emerging Scholars in Engineering,* Vanderbilt University

2021 *BraiNets Lab,* Institut de Neurosciences de la Timone

2021 *Systems Lab Meeting,* Melbourne University

2021 *Neurostats Oxford Meeting,* Oxford University

2021 *Innovators in Cognitive Neuroscience*. Recording: https://www.youtube.com/watch?v=lm80J8-dbS0

2021 *Cognitive Development Lab,* Columbia University

2020 *Appetitive Science Seminar Series,* Yale University

2017 *Magnetic Resonance Research Center Seminar Series,* Yale University

2015 *Yale Epilepsy Research Retreat 2015,* Yale University

2015 *NeuroDay 2015,* Yale University

Invited Conference Talks & Workshops

2026 Symposium: Career Journeys in Open Science. Speakers: Margulies, D., Gould Van Praag, C., **Noble, S.**, Suppiah., S. *Organization for Human Brain Mapping Meeting: Open Science Room*. Bordeaux, FR.

2025 User-friendly web apps to facilitate fMRI study design and analysis. *Neurohackademy.* Seattle, WA.

2024 Symposium: Reproducibility crisis revisited. Speakers: Chen, Z., **Noble, S.,** Hebart, M. *MRI Together.* Virtual.

2024 User-friendly web apps to facilitate fMRI study design and analysis. *Neurohackademy.* Seattle, WA.

2023 Keynote: Making open science work for you. *Advanced Computational Neuroscience Network (ACNN) Big Data Neuroscience Workshop.* Columbus, Ohio.

2023 Workshop: Making open science work for you: Tools for Reproducible Neuroscience. *Advanced Computational Neuroscience Network (ACNN) Big Data Neuroscience Workshop.* Columbus, Ohio.

2023 Symposium: Paths to increased brain-behavior reproducibility. Speakers: Dosenbach, N., **Noble, S.,** Marek, S., Yeo, T., Poldrack, R. *Cognitive Neuroscience Society Meeting*. San Francisco, CA.

2022 Symposium: MRI & Accessibility. Speakers: de Leener, B., **Noble, S.**, Kolind, S. *TransMedTech Conference Series*. Virtual.

2020 The constrained network-based statistic: A new level of inference for neuroimaging. *NIH BRAIN Initiative Alliance's Tools, Tech, Theory and Trainee Series* and *Neuromatch Conference 3.0*. Virtual.

Contributed Conference Talks, Panels, & Symposia

2026 Symposium: Precision Mental Health with Intensive Longitudinal Data and Machine Learning: From Neuroimaging to Ecological Behavior. *Association for Psychological Science Meeting.* Organizer: Zainal, N.H. Speakers: **Noble, S.**, Curtiss, J., Van Doren, N.

2026 ***(Merit Abstract Award)*** Shearer, H., Rosenblatt, M., Ye, J., … Mejia, A., Nichols, T., Curtiss, J., Scheinost, D., **Noble, S.** What effect sizes can we expect in fMRI? Modeling effect sizes distributions across the brain. *Organization for Human Brain Mapping Meeting.*

2026 Symposium: Balancing Sample Size and Sensitivity: Emerging Methods and Tools for Optimal Inference & Power Analysis. *Organization for Human Brain Mapping Meeting.* Organizers: Mansour, S., Lindquist, M., **Noble, S.**, Maumet, C. Speakers: Shearer, H., Wang, S., Larsen, N.Y.

2025 ***(Selected for Oral Session)*** Shearer, H., **Noble, S.** BrainEffeX: An interactive web application for exploring typical fMRI effect size estimates. *Organization for Human Brain Mapping Meeting.*

2025 ***(Selected for Data Blitz)*** Shearer, H., **Noble, S.** BrainEffeX: A web app for exploring fMRI effect sizes. *Cognitive Neuroscience Society Meeting.*

2025 Curtiss, J., **Noble, S**. A Formalization of an IdioNomo Perspective for Psychological Models. *Boston Precision Brain & Mental Health Hack.*

2024 ***(Abstract Award, Selected for Talk)*** Cravo, F., **Noble, S.** A Preliminary Empirical Web-Based fMRI Power Calculator for Increased Reliability. *MRI Together.*

2024 **Noble, S.,** Shearer, H. Mapping the Landscape of Effect Sizes in fMRI: Insights from Large Publicly Available Datasets. *Brainhack Global: Northeastern.*

2024 Symposium: Increasing International Collaboration Opportunities for Early Career Researchers. *Organization for Human Brain Mapping Meeting.* Organizers: Moran, K., Williams, B., Chen, J., Srisaikaew, P. Speakers: Milham, M., **Noble, S.**, Shine, M.

2024 Curtiss, J.E., **Noble, S.** Invited talk: Precision mental and cognitive health using time-series and neuroimaging methodology. *Northeastern University Life Sciences Day*.

2023 Symposium: Advances in Individual-Level Modeling. *Organization for Human Brain Mapping Meeting.* Speakers: Mejia, M. (organizer), **Noble, S.** (organizer), Chen, G. (organizer), Chang, C., Xue, A.

2023 Symposium: Inference on the Brain: Advances and Practices in Brain Activity Inference. *Organization for Human Brain Mapping Meeting.* Organizers: Mansour, S., Zalesky, A. Speakers: Larivière, S., Mansour, S., **Noble, S.**, Taylor, P.

2022 Panel: Emerging Topics in Promoting Reproducible Research from a Statistical Perspective. *Organization for Human Brain Mapping Meeting: Open Science Room.* Organizers & moderators: **Noble, S.**, Bayer, J. Speakers: Mejia, A., Thirion, B., Chang, C., Chen, G., Thompson, W.

2022 Symposium: The Ups and Downs of Open Science: Perspectives from Early-Career and Established Researchers. *Organization for Human Brain Mapping Meeting.* Organizers: de Leener, B., Bayer, J., Moia, S., Parkes, L. Speakers: de Leener, B., Suppiah, P., Gould van Praag, C., **Noble, S.**

2022 RoundTable Discussion: Best Practices for Promoting Diversity and Inclusivity Across OHBM Organizations. *Organization for Human Brain Mapping Meeting.* OSSIG Panelists: **Noble, S.,** Bottenhorn, K., and Bayer, J.

2021 Panel: Aperture and Open Science Roundtable. *Organization for Human Brain Mapping Meeting.* Moderators: Nikolaidis, A., Poline, J.B. Panelists: **Noble, S.,** Lipp, I.

2021 Panel: Ensuring Open Science Is Accessible. *Organization for Human Brain Mapping Meeting: Open Science Room.* Moderator: **Noble, S.** Speakers: Klusza, S., Nolan, S., Klinger, A., Paparella, A.

2021 Symposium: Current Frontiers in Statistical Inference for Neuroimaging Data. *Organization for Human Brain Mapping Meeting.* Organizer: Thirion, B. Moderator: Mumford, J. Speakers: **Noble, S.**, Rosenblatt, J.D.

2021 Symposium: Functional Networks. *IEEE International Symposium on Biomedical Imaging (ISBI).* Co-Organizers: Preti, M.G., Karahanoglu, I. Speakers: Bassett, D., **Noble, S.**, Chen, J., Preti, M.G., Cabral, J.

2021 **Noble, S.,** Scheinost, D. Lightning talk: Leveling Up: Improving power in functional connectivity by moving beyond cluster-level inference. *Writing Your Own Blueprint: The NIH Blueprint Diversity Conference.*

2020 **Noble, S.,** Scheinost, D. Oral Session: The constrained network based statistic: A new level of inference for neuroimaging. *Medical Image Computing and Computer Assisted Intervention (MICCAI).*

2020 **Noble, S.,** Dadashkarimi, J., Saltzman, Z., Lacadie, C., Garbus, H., Casetti, D., Onofrey, J., Papademetris, X., Scheinost, D. Tutorial: Try BioImage Suite Web, a modern and powerful software for neuroscience. *Brainhack New York 2020.*

2020 **Noble, S.,** Dadashkarimi, J., Papademetris, X., Scheinost, D. Talk & Demo. Web native data analysis with WebAssembly: a BISWeb demo and conversation. *Organization for Human Brain Mapping Meeting: Open Science Room.* Recording: https://www.youtube.com/watch?v=9Xgn7Jg7ypo

2020 Symposium: Measuring the Individual: Understanding Sources of Variability in Task and Resting fMRI. *Organization for Human Brain Mapping Meeting.* Organizer: Hawco, C. Speakers: **Noble, S.**, Dickie, E., Gratton, C., Hawco, C.

2020 Dadashkarimi, J., **Noble, S.,** Greene, A., Constable, R.T., Papademetris, X., Scheinost, D. Software demo. On Visualization and Interpretation of Complex Connectomic Results. *Organization for Human Brain Mapping Meeting.*

2019 ***(Merit Abstract Award)*** **Noble, S.,** Scheinost, D., Constable, R.T. Oral Session. Cluster Failure or Power Failure? Evaluating Sensitivity in Cluster-Level Inference. *Organization for Human Brain Mapping Meeting.* Recording: https://www.pathlms.com/ohbm/courses/12238/sections/15843/video_presentations/138325

2019 **Noble, S.,** Dadashkarimi, J., Saltzman, Z., Lacadie, C., Garbus, H., Casetti, D., Onofrey, J., Papademetris, X., Scheinost, D. Talk & Demo. Introducing BioImage Suite Web. *Organization for Human Brain Mapping Meeting: Open Science Room.*

2019 Symposium: Towards Understanding Individual Variability with Functional Neuroimaging: Big Data and Deep Data Perspectives. *Cognitive Neuroscience Society Meeting.* Organizers: Hawco, C., Gratton, C. Speakers: **Noble, S.**, Gratton, C., Hawco, C., Shine, M.

2019 **Noble, S.*,** Saltzman, Z.*, Dadashkarimi, J., Lacadie, C., Garbus, H., Casetti, D., Onofrey, J., Papademetris, X., Scheinost, D. Tutorial. Introducing BioImage Suite Web. *Brainhack Yale 2019*.

2018 **Noble, S.*,** O'Connor, D*. Tutorial. Intro to Machine Learning for fMRI with Nilearn. *Brainhack Yale 2018*.

Select Posters

2026 Fischbach, A., Shearer, H., Handwerker, D., Lewis, L., Quigley, K.S., Theriault, J.E., Satpute, A.B., Feldman Barrett, L., **Noble, S**. Deconfounding Subcortical Signals in fMRI by Modeling Region-Specific CSF Noise. *Organization for Human Brain Mapping Meeting.*

2026 ***(Postdoctoral Award)*** Cravo, F., Fischbach, A., Shearer, H., **Noble S**. PRISME: A MATLAB Toolbox for Multi-Method Statistical Power Analysis in Neuroimaging. *Cognitive Neuroscience Society Meeting.*

2026 Shearer, H., Rosenblatt, M., Ye, J., … Mejia, A., Nichols, T., Curtiss, J., Scheinost, D., **Noble, S.** What effect sizes can we expect in fMRI? Modeling effect sizes distributions across the brain. *Organization for Human Brain Mapping Meeting.*

2026 Olafsson, V., **Noble, S.,** Reid, B., Gabard-Durnam, L., Peelle, J. Preliminary examination of the test-retest stability of subcortical iron imaging via R2'. *Organization for Human Brain Mapping Meeting.*

2024 Jones K.G., **Noble, S.,** Gilbert, J., Yen, M., and Welsh, R.C. Test-Retest Reliability of Magnetic Resonance Imaging Data Across the Siemens VE11C to XA30 Software Upgrade. *International Society for Magnetic Resonance in Medicine (ISMRM).*

2021 **Noble, S.**, Dadashkarimi, J., Saltzman, Z., Lacadie, C., Garbus, H., Onofrey, J., Papademetris, X., Scheinost, D. Bioimage Suite Web: A Simple, Modern, & Powerful Software Suite. *International Neuroinformatics Coordinating Facility (INCF) Assembly.*

2015 ***(Best Poster Award)*** **Noble, S.**, Scheinost, D., Cannon, T.D., Constable, R.T. Reliability of Multisite Functional Connectivity. *Yale Biomedical Imaging Research Retreat.*

Industry Presentations

2013 **Noble, S.**, Poeuv, S., Brewer, J.A. Demonstration for undisclosed media outlet.

2013 Poeuv, S., **Noble, S.**, Pal, P., Brewer, J.A. YEI Innovation Fund. Yale University.

2013 Poeuv, S., **Noble, S.**, Brewer, J.A. CI Pre-Seed Program. Connecticut Innovations, Rocky Hill, CT.

2012 **Noble, S.**, Poeuv, S., Brewer, J.A. TechStart Demo Day. Yale University.

2012 **Noble, S.**, Poeuv, S., Brewer, J.A. Demonstration for Professional Golfer's Association (PGA): Metropolitan Section. Metropolitan PGA Golf Central Offices, Elmsford, NY.

2012 **Noble, S.**, Poeuv, S., Brewer, J.A. Demonstration for New Haven Independent. goBlue Labs.

2012 Poeuv, S., **Noble, S.**, Brewer, J.A. New Haven Start-up Competition. Yale University (anonymous investor).

2012 Poeuv, S., **Noble, S.**, Brewer, J.A. TechStart Demo Day. Yale University.

2012 Poeuv, S., **Noble, S.**, Brewer, J.A. TechStart Accelerator Competition. Connecticut Innovations.

Mentorship

(Affiliation = Northeastern University unless otherwise stated)

Postdoctoral Researchers

2024– Fabricio Cravo

Doctoral Students

2025– Hallee Shearer (Psychology)
2023– Alexandra Fischbach (Psychology)

Research Staff

2023–25 Hallee Shearer

Doctoral Thesis Committees

2024– Jinyu Wang (Interdisciplinary Design and Media; Advisor: Psyche Loui)
2023– Chris Camp (Yale University Interdepartmental Neuroscience Program; Advisor: Dustin Scheinost)
2024–26 Alex Fischbach (Psychology; Advisor: Stephanie Noble)
2025–26 Claire Shaffer (Psychology; Advisors: Susan Whitfield-Gabrieli & Lisa Feldman-Barrett)
2025–26 Trevor Kline (Psychology; Advisors: Aaron Seitz & Susanne Jaeggi)
2024–26 Emma Tinney (Psychology; Advisors: Charles Hillman & Timothy Morris)

Qualifying Exam Committees

2026 Yiyi Liu (Bioengineering; Advisor: Qianqian Fang)

Fellowship Mentorship Teams

2024– Luis Ramirez (UCSD Psychology; Advisor: John Serences): NIH DSPAN F99/K00
2024–25 Goretti España-Irla (Physical Therapy; Advisor: Timothy Morris): CBH Postdoctoral Fellowship
2024–25 Emma Margolis (Psychology; Advisor: Laurel Gabard-Durnam): CBH Graduate Student Fellowship
2024–25 Hailey Smith (Psychology; Advisor: Jonathan Peelle): CBH Graduate Student Fellowship

Visiting PhD Students

2024– Raimundo Rodriguez (Yale University Interdepartmental Neuroscience Program)

Masters Thesis Advisor

2025–26 Chan Cha Hong (Bioengineering)

Co-op Program Adviser

Spring 2024 Catherine Cahill (Bioengineering)

Undergraduate Capstone Advisor

2023–24 Jennifer Field (Mathematics & Physics)

Volunteers & Interns

2025– Ngozi Nnaeto (Postbaccalaureate)
2024– Fatemeh Doshvargar (Postbaccalaureate)
2024 Aneesh Baskaran (Masters student)
2024 Samuel Joseph (Undergraduate student)
2018–19 Tracy Lu (High school student)
2016 Samantha Steinberg (High school student)

Other Mentorship

2019–23 **Assisted in the supervision of** 8 PhD students and 2 research staff in the MINDS Lab (PI: Dustin Scheinost; Departments: Interdepartmental Neuroscience Program, Biomedical Engineering, Biostatistics, Computer Science; Yale University): Maya Foster, Chris Camp, Jean Ye, Raimundo Rodriguez, Matthew Rosenblatt, Wei Dai, Javid Dadashkarimi, Link Tejavibulya, Iris Cheng, and Hannah Petersen

2019–23 **Extracurricular mentorship of** 9 undergraduate and high school students through goBlue and Yale outreach programs (Women in Science at Yale, YBDIC-PATHS, ManyMentors), including Evelyn Soria-Goyzueta (now a BSN graduate, American International College) and Darlis Juvino (now a PhD student, University of Connecticut)

Teaching

UNIVERSITY TEACHING

Creator & Instructor: "PSYC 7250 / BIOE 7374: A Data Science Toolkit for Human Neuroscience Research" *Northeastern University*
Spring 2024, 2025, 2026
PSYCHOLOGY DEPARTMENT (2024-); BIOENGINEERING DEPARTMENT (2025-); 15-20 PHD STUDENTS

Guest Lecture: "Test-retest reliability; Integration and harmonization of multi-site fMRI data" *Harvard-MIT*
2024
HST 583: FMRI: DATA ACQUISITION AND ANALYSIS 2024

Teaching Fellow
ASTR 180: Introduction to Relativity
Yale University
2018

Teaching Fellow
MCDB/NSCI 320a/720a: Neurobiology
Yale University
2015

WORKSHOPS & SHORT COURSES

Guest Lecture: "A guide to the measurement and interpretation of fMRI test-retest reliability"
Training in Advanced Methods in Neuroimaging and Genetics 2022; 2023; 2024; 2025; 2026
University of Utah
Spring 2022, 2023, 2024. 2025, 2026

Lecture (co-presenter): "Reliability and Data Quality"
With Shearer, H., Noble, S., Bray, S., Xu, T., Vanderwal, T.
OHBM Educational Course: "Best Practices in Naturalistic Neuroimaging"
Virtual
2025

Lecture: "Cluster failure or power failure? Empirically evaluating sensitivity and specificity of classical fMRI inference"
OHBM Educational Course: "Beyond blobology: advances in statistical inference for neuroimaging"
Montreal, Canada
2023

Course Organizer: "Cultivating open science practices in academic research and culture"
OHBM Educational Course
Glasgow, Scotland
2022

Workshop: "Try BioImage Suite Web, a modern and powerful software for neuroscience"
Brainhack NY
New York University
2020

Private Tutor: Statistics & Data Science (2 students)
Yale University
2017 – 2020

Workshop: "Introduction to BioImage Suite Web"
Brainhack Yale
Yale University
2019

Workshop: Connectome-based Predictive Modeling Working Group
Magnetic Resonance Research Center
Yale University
2019 (Monthly)

Workshop: "Intro to Machine Learning for fMRI with Nilearn"
Brainhack Yale
Yale University
2018

Ad Hoc Review & Editorial Membership

Web of Science: https://www.webofscience.com/wos/author/rid/AEE-8968-2022

Editorial Board: Imaging Neuroscience, NeuroImage: Reports (2022—2023; NeuroImage editorial board moved to Imaging Neuroscience)

Editorial Review Board: Aperture Neuro (OHBM)

Guest Editor: Developmental Cognitive Neuroscience 2024 Special Issue "Methodological and analytic advances in developmental neuroscience"

Ad Hoc Review
General Science
Proceedings of the National Academy of Sciences (PNAS), Nature Communications, Nature Methods, Nature Scientific Reports, Nature Scientific Data, Nature Communications Biology, Advanced Science, Science Advances, PLOS One, eLife, Cell Reports
Neuroscience & Psychology
Aperture Neuro, Imaging Neuroscience, Human Brain Mapping, Cerebral Cortex, Network Neuroscience, Nature Neuroscience, Nature Human Behavior, Nature Mental Health, Neuron, Journal for Reproducibility in Neuroscience, Journal of Neuroscience Research,

NeuroImage (2016-2023), eNeuro, Social Cognitive and Affective Neuroscience, Psychophysiology, Social Cognitive and Affective Neuroscience, Developmental and Cognitive Neuroscience, Trends in Cognitive Sciences

Computational & Statistical Methods

Advanced Intelligent Systems, PLOS Computational Biology, MICCAI Medical Image Analysis, Multivariate Behavioral Research, IEEE Transactions in Biomedical Engineering, Computers in Biology & Medicine, F1000

Clinical

NeuroImage: Clinical, Psychiatry Research: Neuroimaging, Schizophrenia Bulletin, Behavior Change, Assessment, BMC Psychiatry

Grant Review: Deutsche Forschungsgemeinschaft (DFG)

Academic Leadership & Service

Boston Precision Brain & Mental Health Hack '25 Co-creator & Organizer — *Fall 2025*

Northeastern University Biomedical Imaging Center Search Committee Member — *2025-*

Psychology Social Committee, Northeastern University Co-chair — *2025-*

Institute for Cognitive & Brain Health Social Committee, Northeastern University Co-chair — *2025-*

Pre-Proposal Grant Review, Northeastern University Ad Hoc Reviewer (2 Mechanisms) — *2025-*

College of Engineering PhD Council Wellness Coffee Hour Invited Speaker: "Beyond the Degree: Career Paths, Networking, and Media Presence" — *Summer 2025*

Brainhack Northeastern (Branch of Brainhack Global) '23, '24 Creator & Organizer — *Winter 2024, Fall 2024*

Boston Imaging Interest Group (BIIG) Journal Club Co-creator & Organizer — *2024-*

Psychology Colloquium Committee, Northeastern University Co-chair — *2024-*

Psychologists for Equity and Inclusion (PIE) Committee, Northeastern University — *2024 -*

Diversity, Equity, Inclusivity in Bioengineering (DEIB) Committee, Northeastern University — *2023 -*

Institute for Cognitive & Brain Health Computing Committee, Northeastern University Founder; Co-chair — *2023 -*

YBDIC-PATHS Mentoring Program Mentor — *2020-2021*

Brainhack Yale 2019 Lead Organizer; Workshop Instructor — *Spring 2019*

Annie Le Fellowship Selection Committee, Yale University — *Spring 2019*

INP Diversity Recruitment Panel, Yale University Panelist; SWE Representative — *Spring 2019*

Brainhack Yale 2018: Lead Organizer; Workshop Instructor — *Fall 2018*

Yale Minority Scientists Research Network Board Member — *Spring 2018*

INP Speaker Seminar Committee, Yale University — *Spring 2017*

She Started It, Yale University Invited Panelist: "Women in Entrepreneurship" — *Spring 2017*

McDougal Center, Yale University Communications Assistant — *Spring 2016*

Mind Matters, Yale University Invited Panelist: "Race and Mental Health" — *Spring 2016*

Women in Science at Yale Mentor; "Career Strategy" Panelist ('14-'16) — *2014-2018*

Yale Graduate Society of Women Engineers Outreach Chair ('15-'17); Mentor; Volunteer; Panelist — *2014-2017*
Led four outreach events and two networking/career building events (panelist)

INP Outreach Committee, Yale University Chair ('15-16); Volunteer ('14-17); Speaker ('16, '18 NIH BP-Endure) — *2014-2016*
Led six outreach events per year (30-60 students per event)

La Casa Cultural, Yale University Mentor — *2014-2015*

Yale Office for Graduate Student Development and Diversity Mentor — *2014-2017*

Yale Graduate Visual Artists Society Founder ('14); Organizer — *2014-2016*

ManyMentors / New Haven Science Fair, Yale University Mentor — *2014-2015*

Princeton Biomedical Engineering Society President ('11-'12), VP ('10-'11), Cofounder *2010-2012*

External Leadership & Service

Brainhack Global 2025 Web Volunteer *2025*

Columbia University POR Colloquium Invited Talk: "Career Paths, Networking, & Social Media Presence" *Fall 2025*

Neurohackademy Invited Panelist: "Careers in neuroimaging and data science" (two separate years) *Summer 2024, 2025*

NIH DSPAN Annual Meeting 2024 Invited Panelist: "Navigating the road to faculty" *Fall 2024*

NIH DSPAN BrainTrust 2023 Invited Panelist: "Mechanisms towards Independence" *Fall 2023*

OHBM Naturalistic Imaging Best Practices Working Group Member (Chairs: Tamara Vanderwal & Einat Liebenthal) *2024-*

Brainhack Global 2023 Volunteer; Outreach Team *2023*

Harvard Med / MGH Postdoc Fellowship Seminar Invited Speaker: "Having an online presence" *Fall 2022, 2023*

NIH Blueprint-ENDURE 2022 Invited Panelist: "Pathways and perspectives on advancing your career" *Fall 2022*

WINRepo Chat 2022 Invited Moderator: "Grants & Fellowships" (prof dev program created by Vale Borghesani) *Fall 2022*

Brainhack Global 2022 Diversity, Equity, & Inclusion Team Lead; Outreach Team; Onboarding Team *Fall 2022*

OHBM OSSIG 2022 Table Talks Facilitator: "Reproducible science and my role in it (Thomas Nichols)" *Summer 2022*

OHBM OSSIG 2022 Poster Pals Organizer (poster networking program created by Sarah Goodale) *Summer 2022*

WINRepo Volunteer *2021-present*

OHBM Open Science Special Interest Group (OSSIG) 2022 Inclusivity Officer *2021-2022*
Co-organized Open Science Room panels and events

OHBM Diversity & Inclusivity Committee (DIC) 2022 OSSIG-DIC Liason *2021-2022*

Científico Latino Graduate School Mentorship Initiative Application Reviewer (2 students) *Fall 2021*

Brainhack Global 2021 Diversity, Equity, & Inclusion Team Lead; Outreach Team; Onboarding Team *Fall 2021*

Brainhack OHBM 2021 Brainhack Diversity, Equity, & Justice Team Lead; Social Media Lead; Mentor *Spring 2021*

Brainhack Global 2020 Social Lead Organizer *Fall 2020*

Neuromatch Conference 3.0 Moderator (4 traditional symposia, 1 interactive symposium) *Fall 2020*

Columbia University POR Colloquium Invited Talk: Grant Funding Seminar *Fall 2020*

FIT'NG Flux Preconference Workshop Moderator: "Data Sharing" Breakout Session *Fall 2020*

OHBM 2020 Club Night Social Lead Organizer *Summer 2020*

NIH Blueprint D-SPAN F99/K00 Webinar Invited Panelist *Winter 2019*
https://www.ninds.nih.gov/News-Events/Events-Proceedings/Events/NIH-Blueprint-D-SPAN-F99K00-Webinar

Neuroscience Scholars Program Leadership Meeting Invited Panelist *Summer 2019*

Brainhack Networks 2019 Invited "Team of Experts" *Winter 2019*

NIH Blueprint D-SPAN F99/K00 Twitter Q&A Panelist *Fall 2018*

Neuroscience Scholars Program Leadership Meeting Invited Member *Summer 2017*

Connectionism Art Movement Founder; Organizer *2012-2014*

Open Science Contributions

Selected contributions (for full list, see https://github.com/neuroprismlab?tab=repositories)

BrainEffeX *code* *2024*

https://neuroprismlab.shinyapps.io/effect_size_shiny/

PRISME Empirical Power Curve Generation Toolbox *code* *2020*

https://github.com/neuroprismlab/PRISME-Brain-Power-Calculator

Network-Based Inferential Procedures and Benchmarking Toolbox (including cNBS) *code* *2019*

https://github.com/neuroprismlab/NBS_benchmarking

Cluster-Based Inference Benchmarking Toolbox *data* *2018*

https://github.com/sneuroble/cluster_power_failure

Yale Test-Retest Dataset *code* *2018*

http://fcon_1000.projects.nitrc.org/indi/retro/yale_trt.html

Multifactor ICC Toolbox

https://github.com/neuroprismlab/Multifactor_ICC

Professional Memberships

Organization for Human Brain Mapping (2015–present)
Cognitive Neuroscience Society (2019, 2022—present)
Association for Psychological Science (2024)
Association for Behavioral and Cognitive Therapy (2023)
Society for Neuroscience (2014–2022)
International Symposium on Biomedical Imaging (2021)
Medical Image Computing and Computer Assisted Intervention (2020)

Skills

Programming Matlab, R, bash, Python, C++, HTML5

Art & Design Graphic design & traditional media (watercolor, oil, pastel) including logos for BioImage Suite Web, FIT'NG